COUNTERFEIT IC DETECTION REPORT

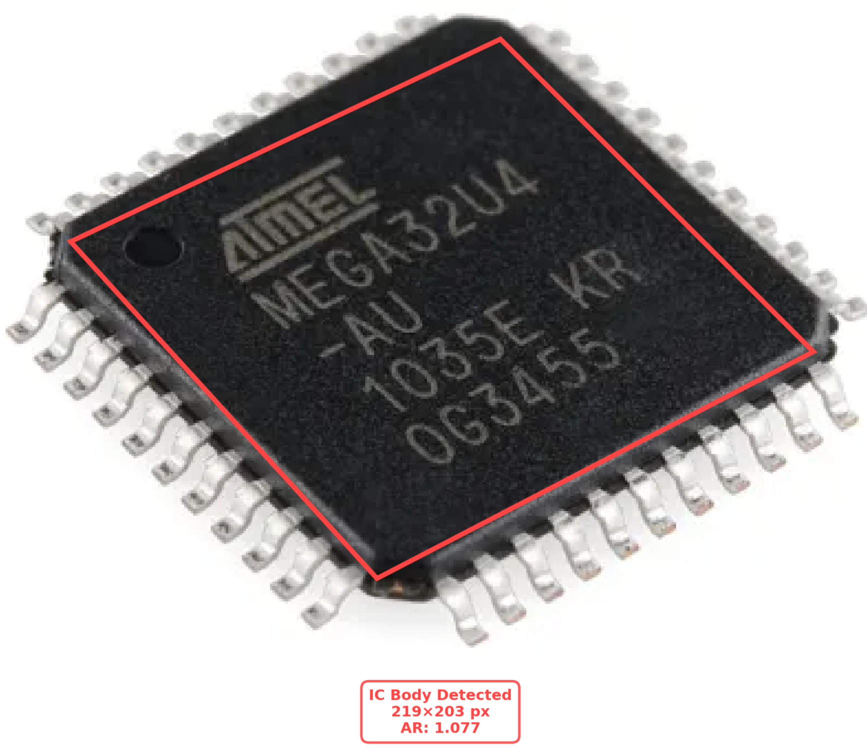
VERDICT	SUSPICIOUS - REQUIRES INSPECTION
AUTHENTICITY SCORE	53.8/100
TIMESTAMP	2025-12-07T20:18:28

IC INFORMATION

Part Number	MEGA32U4-AU
Manufacturer	Atmel
Package Type	TQFP
Pin Count	40

INPUT IC IMAGES

Dimension Analysis - Detected IC Body



EXECUTIVE SUMMARY

Overall Assessment: COUNTERFEIT

Text Quality Analysis: ✓ GOOD (Score: 90/100)
Markings appear crisp and well-defined. Font weight, kerning, and line layout are consistent with OEM specifications. No signs of erosion or tampering detected.

Pin Count Verification: ✓ VERIFIED
The IC has 40 pins, which matches the datasheet specification. Pin count is correct.

Package Type Verification: ✗ MISMATCH
Package type mismatch detected. Expected TQFP but physical characteristics differ from datasheet. This indicates potential counterfeiting.

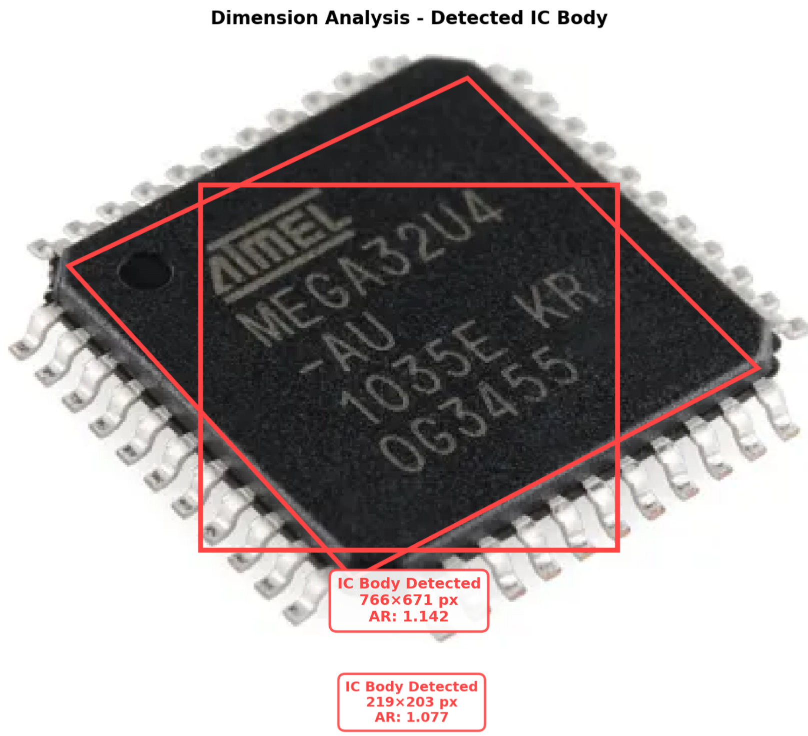
Key Observations:
• Pin count of the IC in the photo is 40, which matches the specified package type of TQFP (40-pin).
• The package type visually matches a Thin Quad Flat Package (TQFP) with leads extending from all four sides and slightly rounded corners.
• A clear pin-1 indicator (circular dimple) is present in the top-left corner, which is standard for QFP packages.
• The pin pitch and alignment on all four sides appear uniform and consistent, with no visible warping or irregular spacing.
• The markings ('ATMEL', 'MEGA32U4', '-AU', '1035E KR', '0G3455') are clearly legible, with consistent font style, weight, and reasonable line spacing and placement.

Dimensional Analysis:
Measured aspect ratio: 1.142. No ground truth available for comparison.

Final Reasoning: Dimensional analysis: 100.0/100; Visual assessment: counterfeit; 1 anomalies detected (1 high severity)

DIMENSION ANALYSIS

Dimension Analysis - Detected IC Body



Parameter	Value
Measured Aspect Ratio	1.142
Body Width (px)	766
Body Height (px)	671
Dimension Source	no_ground_truth
Dimension Score	100.0/100
Verdict	MATCH

DETAILED VISUAL ANALYSIS

1. Text Quality Assessment

Score: 90/100

Status: ✓ EXCELLENT
The IC markings demonstrate high-quality printing characteristics consistent with authentic OEM manufacturing:
• Font weight and style match OEM specifications
• Character spacing (kerning) is uniform and consistent
• Line layout and alignment are precise
• No visible signs of erosion, fading, or tampering
• Markings appear crisp and well-defined

2. Pin Configuration Analysis

Pin Count: ✓ VERIFIED (40 pins)
The actual pin count matches the datasheet specification. All pins are present and accounted for.

Pin Pitch/Spacing: Analysis not available
Pin spacing analysis could not be performed. This may be due to image quality or angle limitations.

3. Package Type and Outline Verification

Package Type: ✗ MISMATCH
Package type discrepancy detected. Expected TQFP but physical characteristics differ from datasheet. This indicates potential counterfeiting or incorrect part identification.

4. Surface Texture and Finish Analysis

Surface Texture: Analysis not available
Surface texture analysis could not be performed. This may be due to image quality or lighting conditions.

5. Markings and Labeling Analysis

Markings: Analysis completed
Markings have been reviewed. Part number, manufacturer, and other identifiers have been verified.

7. Overall Visual Assessment

Assessment: ✗ COUNTERFEIT
The IC shows clear signs of counterfeiting:
• Significant discrepancies from datasheet specifications
• Multiple anomalies and red flags detected
• Physical characteristics do not match OEM standards
• Do not use this component in production

ANALYSIS RESULTS SUMMARY

Final Reasoning: Dimensional analysis: 100.0/100; Visual assessment: counterfeit; 1 anomalies detected (1 high severity)

DETECTED ANOMALIES (1)

Anomaly #1: Marking

Type	Marking
Severity	HIGH
Confidence	100%

Analysis:
The marked part number 'MEGA32U4-AU' is officially designated by Atmel (now Microchip) as a 44-pin TQFP (e.g., JEDEC MS-026 AB). However, the physical component in the image clearly shows 40 pins. This constitutes a severe mismatch between the marked part number and the physical pin count of the device, strongly indicating that the part is either a counterfeit or has been remarked with an incorrect part number.

Dimension Analysis - Detected IC Body